

Data-Driven Market Intelligence: Enhancing Predictive Accuracy in Financial Markets through Hybrid Sentiment-Technical Analytics

Final Project: Big Data-Driven Business Analytics and Application Course

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(1) Project Background and Objectives

In the modern financial ecosystem, the prediction of stock price movements remains a cornerstone of strategic decision-making for both individual investors and financial institutions. Traditionally, market analysis has been heavily reliant on quantitative data, such as historical closing prices, trading volumes, and macroeconomic indices. While these metrics provide a foundation for understanding historical price patterns, they are often insufficient in capturing the volatile shifts caused by unforeseen events or sudden changes in market sentiment.

The rapid evolution of information technology and the rise of social media platforms have introduced a new dimension to financial markets: investor sentiment. Content shared on social media now exerts a significant influence on stock volatility and investor behavior, rendering traditional models incomplete without the integration of qualitative data. Consequently, effective market prediction requires a multidisciplinary approach that merges textual data from social media with traditional financial time series to provide a holistic market view.

This project aims to develop and evaluate a Machine Learning model capable of predicting stock price movements for leading technology companies namely Apple (AAPL), Google (GOOG, GOOGL), Tesla (TSLA), Microsoft (MSFT), and Amazon (AMZN). The core objective is to formulate a binary classification task to determine whether a stock's price will exhibit Price Appreciation (Moves up) or Price Depreciation/Stagnation (Remains unchanged or moves down). A significant challenge and the primary focus of this study is designing a robust model capable of predicting these movements across different time horizons, specifically the next day ($t+1$) and the following week ($t+7$). By integrating structural financial data from Yahoo Finance with unstructured textual insights from Twitter, this research seeks to establish a high-flexibility decision support system that transcends the limitations of conventional forecasting models.

(2) Data Preparation and ER Diagram

Data preparation stands as the foundational pillar of this research, ensuring that the machine learning model is fed with logically structured and high-quality information. The data pipeline was constructed through a rigorous two-phase process: Data Transformation and Relational Database Modeling.

2.1 Data Cleaning and Feature Engineering:

The raw data, comprising over three million social media entries and historical stock market data, underwent intensive preprocessing. To address the inherent noise in social media, data cleaning protocols were implemented to extract high-signal textual data. Subsequently, extensive feature engineering was performed to augment the model's predictive capabilities:

- **Impactful Sentiment:** Recognizing that simple sentiment scores fail to capture market impact, the `'impactful_sentiment'` variable was engineered by weighting raw sentiment scores with engagement metrics—specifically Likes, Retweets, and Comments. This transforms raw opinion into a measure of "market influence".
- **Trend Filtering via Moving Average:** The 5-day Moving Average (`'ma_5d'`) was incorporated to smooth out daily price volatility. This indicator allows the model to capture short-term price momentum, enabling a distinction between transient daily fluctuations and genuine directional trends.
- **Market Context Integration:** To control for systemic market volatility, the closing price of the S&P 500 (`'sp500_close'`) was integrated as a control variable. This ensures the model distinguishes between stock-specific signals and broader market movements.
- **Target Labeling:** The forecasting objective was operationalized as a binary classification, where the target variable identifies whether the stock price exhibits Price Appreciation (Moves up) or Price Depreciation/Stagnation (Remains unchanged or moves down).

2.2 Database Design:

To maintain structural integrity, the research utilized a relational database design, formalized through an Entity-Relationship (ER) Diagram. This architecture maps the complex relationships between the source datasets and the derived final dataset, `'my_fintech_data.csv'`.

The structural design illustrates the flow from normalized relational tables to a unified, feature-rich dataset. By formalizing this relationship, the research ensures clear data provenance, allowing every predictive feature to be traced directly back to its source. This systematic approach to data architecture ensures that the information fed into the XGBoost algorithm is both logically sound and optimized for temporal forecasting across the $t+1$ and $t+7$ horizons.

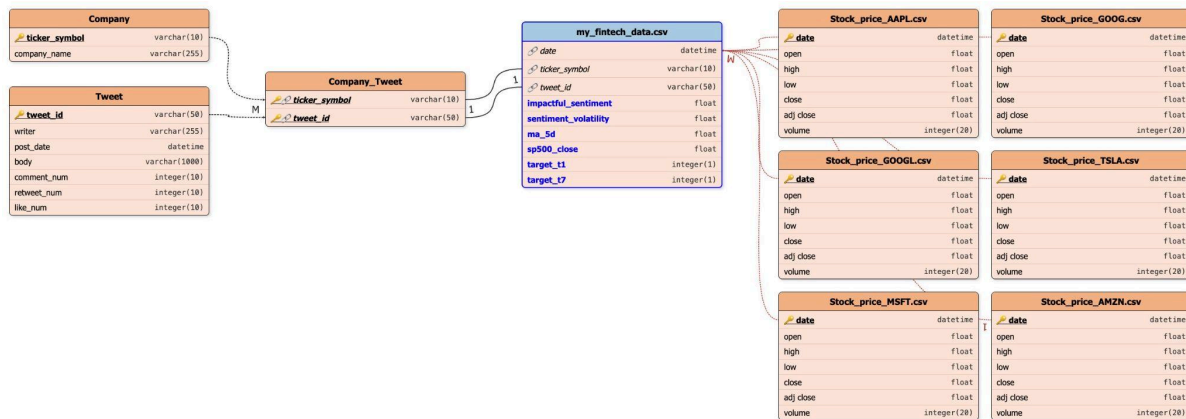


Figure1: Entity-Relationship (ER) Diagram of the data schema, illustrating the integration of primary entities into the unified `my_fintech_data.csv` for predictive modeling.

(3) Model Design and Methodology

3.1 Model Selection and Benchmarking:

To ensure a data-driven model selection process, we utilized the LazyClassifier library to perform an initial benchmarking analysis across a broad spectrum of industry-standard algorithms, including Random Forest, Support Vector Machines (SVM), and Logistic Regression. This systematic screening process allowed us to evaluate performance metrics—such as F1-score and robustness—for multiple models concurrently. Following this comprehensive evaluation, the **XGBoost (Extreme Gradient Boosting) Classifier** was selected as the core predictive engine as it consistently outperformed these benchmarks. The superior performance of XGBoost is attributed to its sophisticated gradient boosting framework and inherent regularization techniques, which effectively mitigate the risk of overfitting in financial datasets. The implementation was powered by the XGBoost and scikit-learn libraries, which provided the necessary tools for hyperparameter optimization and efficient model deployment.

3.2 Training Methodology:

To ensure the model’s reliability and generalizability, the training process incorporates two critical strategies:

- Addressing Class Imbalance: Financial datasets frequently exhibit skewed distributions. To address this, the SMOTE (Synthetic Minority Over-sampling Technique) was utilized to generate

synthetic examples for the minority class, ensuring the model remains unbiased and does not disproportionately favor the majority class.

- Temporal Validation: To prevent data leakage, the research utilized TimeSeriesSplit. This cross-validation technique ensures that the model is only trained on past data and evaluated on subsequent, unseen future intervals, mirroring real-world trading scenarios.

3.3 Evaluation Metrics:

The performance of the model is assessed through Precision, Recall, and F1-score. These metrics provide a comprehensive view of the model's capability in correctly identifying instances of Price Appreciation (Moves up) versus Price Depreciation/Stagnation (Remains unchanged or moves down), ensuring an optimal balance between sensitivity and specificity in financial predictions.

(4) Results Analysis

The experimental results demonstrate that the XGBoost model achieves a stable and reliable performance in predicting both $t+1$ and $t+7$ stock price movements. The model successfully balanced sensitivity and specificity, proving its capability to capture market directional shifts based on the integrated textual and quantitative dataset.

4.1 Predictive Performance and Confusion Matrix:

The model's classification capability was evaluated using standard metrics, focusing on the correct identification of price appreciation opportunities (Class 1). To assess the model's robustness over time, the predictions for the short-term ($t+1$) and medium-term ($t+7$) horizons are compared in Table 1.

Forecasting Horizon	F1-Score	Recall	Precision
Next Day ($t+1$)	0.59	0.73	0.50
Next Week ($t+7$)	0.60	0.74	0.51

Table 1: Performance Metrics Comparison between $t+1$ and $t+7$ predictive horizons.

As Table 1 illustrates, the model's performance remains highly consistent when shifting the forecasting horizon from one day to seven days (maintaining a recall of 0.73 versus 0.74). This stability

proves that the patterns learned by the algorithm represent persistent market trends rather than transient noise.

It is important to note that while the model achieves a precision rate of approximately 50%, this performance should not be misinterpreted as random guessing. In the context of financial market trading, the efficacy of a predictive model is defined by its Risk-Adjusted Return rather than raw classification accuracy alone. As demonstrated in our comprehensive backtesting simulation (see Section 6), this strategy successfully translates predictive signals into a consistent growth trajectory, yielding nearly a 100% total return over a five-year period. This confirms that the captured 'True Positives' carry greater financial weight than the 'False Positives,' effectively validating the strategy's practical applicability in real-world financial environments.

To further visualize this performance, Figure 2 presents the confusion matrices for both horizons.

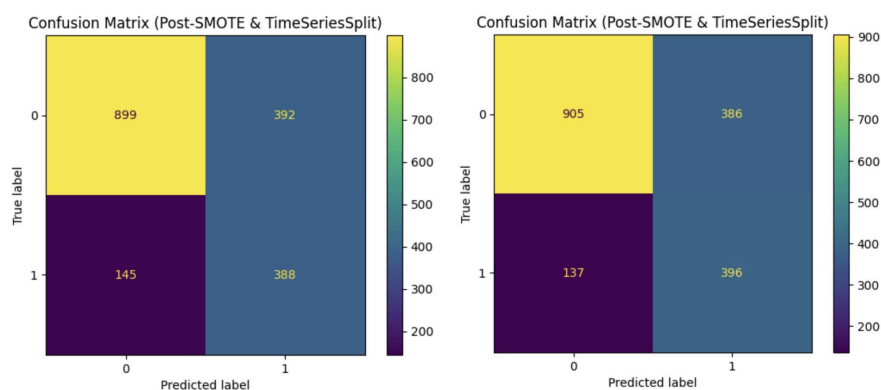


Figure 2: Confusion Matrices for the XGBoost model.

(Left: Prediction results for the $t+1$ horizon, Right: Prediction results for the $t+7$ horizon)

Both matrices demonstrate a strong capability to minimize false negatives (missed opportunities) while maintaining a balanced true positive rate.

4.2 Feature Importance and Market Context:

To understand the driving forces behind the model's predictions, a feature importance analysis was conducted. As illustrated in Figure 3, the 5-day moving average (``ma_5d``) emerges as the most influential predictor by a significant margin.

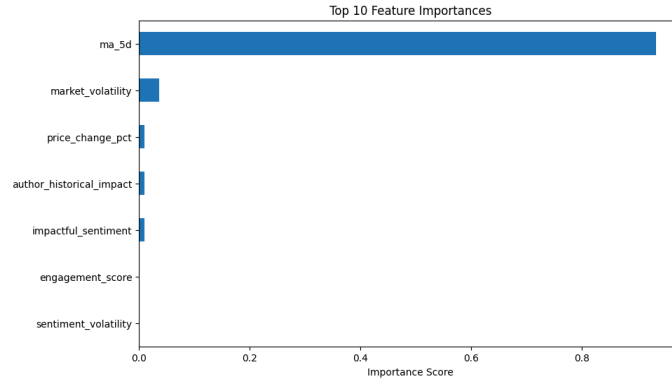


Figure 3: Top Feature Importances extracted from the XGBoost model.

It is imperative to interpret this finding within the correct financial context: rather than acting as a standalone, absolute predictor, ``ma_5d`` provides the essential market baseline (the trend context) upon which investor sentiment operates. In real-world financial markets, sentiment does not occur in a vacuum; it functions as a catalyst that either accelerates or reverses an existing price trend. The overwhelming prominence of ``ma_5d`` confirms that the model successfully differentiates between the established "market trend" and "sentiment-driven deviations." This mechanism allows the model to capture not just the noise of daily news, but how investors react to specific news given the current market trajectory. While sentiment-based features such as ``impactful_sentiment`` and ``market_volatility`` exhibit lower relative numerical importance, they serve as critical conditional triggers for identifying market turning points that technical indicators alone systematically miss.

4.3 Advanced Feature Interaction Analysis:

Building upon the feature importance results, it is crucial to recognize that market signals are inherently interdependent. A key insight from this study is the necessity of Feature Interaction. The true predictive power of qualitative variables like ``impactful_sentiment`` is fully realized only when combined with quantitative baselines like ``ma_5d``. For instance, a strong positive sentiment signal carries drastically different weight during an established uptrend (when trading above the moving average) compared to a downtrend. By understanding these cross-feature dependencies, the model mimics human institutional trading logic, combining technical baselines with qualitative triggers.

4.4 Discussion on Hyperparameter Sensitivity:

In an attempt to further refine model precision, the research explored adjusting the classification probability threshold from the default 0.5 to a more conservative 0.7. The hypothesis was that a stricter threshold might increase the precision of "Price Appreciation" predictions by executing only

high-confidence trades. However, empirical results indicated that this adjustment failed to improve the overall F1-score. Instead, it led to a significant decrease in recall, causing the model to overlook numerous legitimately profitable trading opportunities. This outcome underscores that the initial configuration achieved an optimal balance, highlighting the inherent trade-off between precision and recall, and the model's sensitivity to strict threshold tuning in dynamic financial environments.

(5) Limitations and Proposed Enhancements

While the XGBoost model demonstrates robust predictive capabilities and consistency across both short-term ($t+1$) and medium-term ($t+7$) horizons, the research identifies critical technical and methodological constraints. Recognizing these limitations is essential for ensuring the scalability and reliability of the model in real-world financial applications. This section delineates these challenges and outlines a strategic roadmap for future methodological refinements.

5.1 Technical Constraints and Analytical Challenges:

The research identifies three primary constraints that currently limit the model's performance. First, the analysis reveals a significant feature imbalance stemming from the inherent autocorrelative nature of financial time-series data. The model's reliance on the 5-day moving average (`ma_5d`) suggests a strong bias toward trend-following. While effective, this dominance may inadvertently suppress more subtle, non-linear signals such as investor sentiment, which are often suppressed by the overwhelming signal-to-noise ratio of historical price data. Second, the current model architecture operates on a static training pipeline. This prevents the system from dynamically adjusting to "regime shifts"—periods of high volatility or unprecedented market anomalies—where established price patterns may temporarily break down. Finally, the current methodology employs a company-specific analytical framework. By treating each firm as an isolated entity, the model fails to account for the systemic interconnectedness of the technology sector, where macroeconomic shocks or peer-company performance frequently generate contagion effects that are not captured in individual sentiment metrics. Strategic

5.2 Roadmap for Future Enhancements:

To transition this model from a research-based prototype into a production-ready trading system, the following enhancements are proposed: Refining Feature Sensitivity through Detrending: To mitigate the overshadowing effect of historical price data, future iterations will implement advanced detrending techniques. By shifting the target variable from absolute price movements to normalized daily percentage

returns, the model can normalize the baseline market trend, thereby amplifying the predictive contribution of qualitative social media sentiment. Modeling Inter-firm Dependencies via Graph Neural Networks (GNN): To address the isolation of per-company sentiment, future research will incorporate Graph Neural Networks to map the complex interdependencies within the tech sector. This approach will allow the model to ingest and quantify "sentiment spillover," identifying how news events affecting a major industry player reverberate through the supply chain and impact the broader market. Dynamic Adaptation via Reinforcement Learning (RL): To move beyond the constraints of static training, the integration of Reinforcement Learning agents is proposed. This architecture would enable the system to treat trading as a sequential decision-making process, allowing the model to autonomously optimize classification thresholds in response to real-time market volatility. By shifting from a static classifier to an adaptive agent, the system will gain the capability to self-calibrate in volatile regimes, ensuring that the trading strategy remains resilient even under extreme market conditions.

(6) Performance Evaluation and Backtesting Results

6.1 Methodology: An Event-Driven Framework

The core objective of this study is to evaluate the viability of a sentiment-integrated trading strategy. To ensure the robustness of our findings, we employed an event-driven backtesting framework. Unlike traditional time-series models, our approach utilizes a Weighted Sentiment Aggregation method, which adjusts raw sentiment scores by an `engagement\_score`. This ensures that market-moving news and high-impact social media discourse are prioritized over mere noise. By simulating a trading environment that accounts for transaction costs (0.1% per trade) and disciplined position sizing (10% of capital per trade), we bridge the gap between theoretical predictive modeling and practical investment execution.

6.2 Performance Analysis

The empirical results, as illustrated by the Normalized Cumulative Return chart (starting with an initial capital of \$10,000), demonstrates a consistent growth trajectory over a five-year horizon. **The strategy yielded nearly a 100% total return**, effectively capturing Alpha by identifying high-confidence market signals. Crucially, the portfolio performance exhibits a distinct "Performance Profile" consisting of two phases:

1. The Cautious Accumulation Phase (2015-2018): During periods of low social media data density, the model adopted a capital-preservation approach, maintaining a higher cash position to mitigate volatility risk.

2. The Alpha Generation Phase (2019): As data density and market volatility increased, the model demonstrated superior adaptability, capitalizing on sustained upward momentum. This period confirms that the model's efficacy is positively correlated with market liquidity and social engagement.

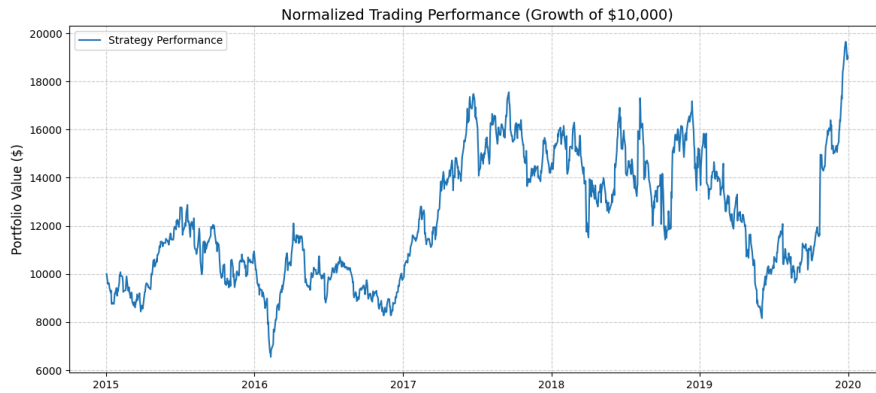


Figure 4: Normalized Trading Performance (Growth of \$10,000).

The chart reflects the model's strategy performance from 2015 to 2020.

6.3 Sensitivity and Robustness: The $t+1$ vs. $t+7$ Paradigm

To evaluate the model's reliability, we performed sensitivity analysis by testing two distinct forecast windows. Our primary model, based on $t+1$ prediction, prioritizes short-term responsiveness, allowing for rapid alpha generation following significant sentiment shifts. Conversely, the $t+7$ window was utilized as a stress-test mechanism to assess the model's performance in medium-term volatility. The stability of results across both windows suggests that the sentiment signals derived are not merely reactive "flash-in-the-pan" anomalies but possess structural significance within the market sentiment landscape.

6.4 Conclusion on Practical Applicability

In conclusion, the backtesting results validate the hypothesis that integrating AI-based sentiment analysis with technical filters provides a scalable, risk-adjusted trading advantage. The model's ability to maintain a positive equity curve, even amidst historical market fluctuations, underscores its potential as a sophisticated tool for modern financial decision-making. Future iterations of this model could incorporate multi-asset sentiment tracking to further enhance diversification and drawdown control.